Statistical self-supervised methods to solve linear inverse problem and application to PET reconstruction.

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March 10, 2026

Table of contents

1. Context
2. Noise2Noise : restore image without clean data
3. Applications

Doctoral project goals

Positron Emission Tomography (PET) imaging

- **PET** is a medical imaging technique that detects pairs of gamma rays emitted by a tracer.
- The patient is injected with a radioactive tracer that emits positrons.
- When a positron meets an electron, they annihilate, producing two gamma photons detected by a ring of detectors.
- The data collected is used to reconstruct images of tracer concentration in the body.

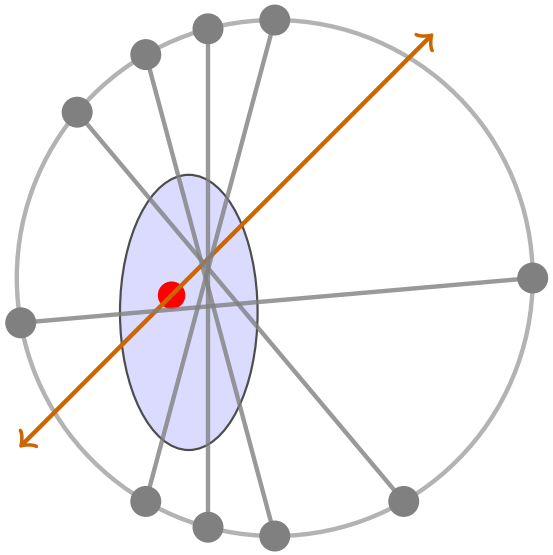
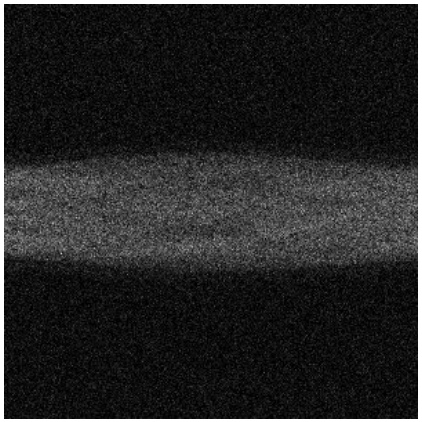


Figure: PET : Emission event detected by ring detectors

PET reconstruction as a linear inverse problem

Context



$= \text{Poisson}\left(A \cdot \right.$ $\left. \right)$

- The PET reconstruction problem can be formulated as a Poisson linear inverse problem :

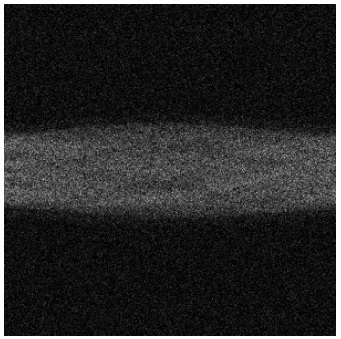
$$Y|X = x \sim \text{Poisson}(Ax) \tag{1}$$

- A is **known** and is the **system/projection matrix**.

PET : Some SOTA reconstruction methods

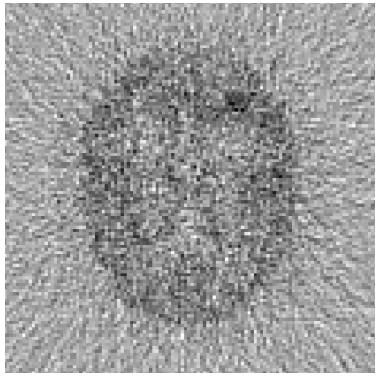
Context

- Iterative statistical reconstruction methods
  - Maximum Likelihood-Based (e.g. MLEM, OSEM)
  - Penalized Likelihood-Based (e.g. MAPEM)
- Analytical reconstruction methods (e.g. FBP)
- Machine Learning / Deep Learning methods
  - Post-Reconstruction enhancement (e.g. CNN denoising)
  - Direct reconstruction (e.g. end-to-end CNN)



Reconstruction (FBP)

↓



PET : ground truth unavailability

Context

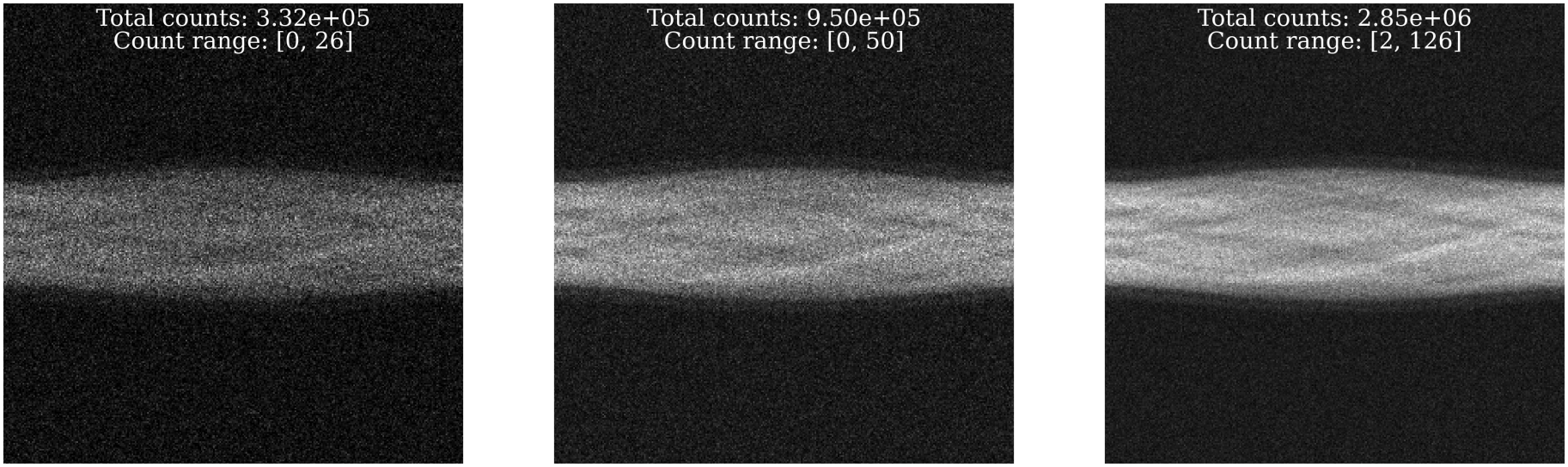


Figure: PET : Noisy prompt sinograms with different count levels

- Raw data is noisy and reference high-count sinograms (right) are less available.
- Ground truth images are not available at all.

The classical supervised framework

Noise2Noise : restore image without clean data

- x is known ; samples are pairs of **noisy** measurement and **clean** image (y, x).
- We want to learn a mapping $f_\theta(y) \approx x$.

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^{N} \rho \left(f_\theta \left(y_i \right) - x_i \right) \tag{2}$$

- Lack of ground truth data, especially in the medical field.

The Noise2Noise insight

Noise2Noise : restore image without clean data

- Samples are pairs of **noisy** measurements $(y^{(1)}, y^{(2)})$ of the same **unknown** underlying image x.
- We want to learn a mapping $Af_\theta(y^{(1)}) \approx y^{(2)}$ [Xia et al.], hence the following objective function :

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^{N} \rho \left(Af_\theta \left(y_i^{(1)} \right) - y_i^{(2)} \right) \tag{3}$$

- Note that $A = \mathrm{Id}$ is the original Noise2Noise formulation [Lehtinen et al. 2018] and is pure denoising.

Statistical conditions

Noise2Noise : restore image without clean data

- We will consider the following statistical conditions :

$$\begin{cases} \mathbb{E}\left[Y_i | X = x\right] = Ax & \text{(centered noise)} \\ Y_1 | X \perp\!\!\!\perp Y_2 | X & \text{(independancy)} \end{cases} \tag{4}$$

Proposition 2.1

Under the statistical conditions 4, the optimal solution of the Noise2Noise loss 3 is the same as the optimal solution of the supervised loss for some good choice of ρ as $N \to \infty$.

Statistical conditions

Noise2Noise : restore image without clean data

Proposition 2.2

If ρ is the L_2 loss, then proposition 2.1 is satisfied and the optimal solution is the ***mean*** *of noise realisations.*

Proposition 2.3

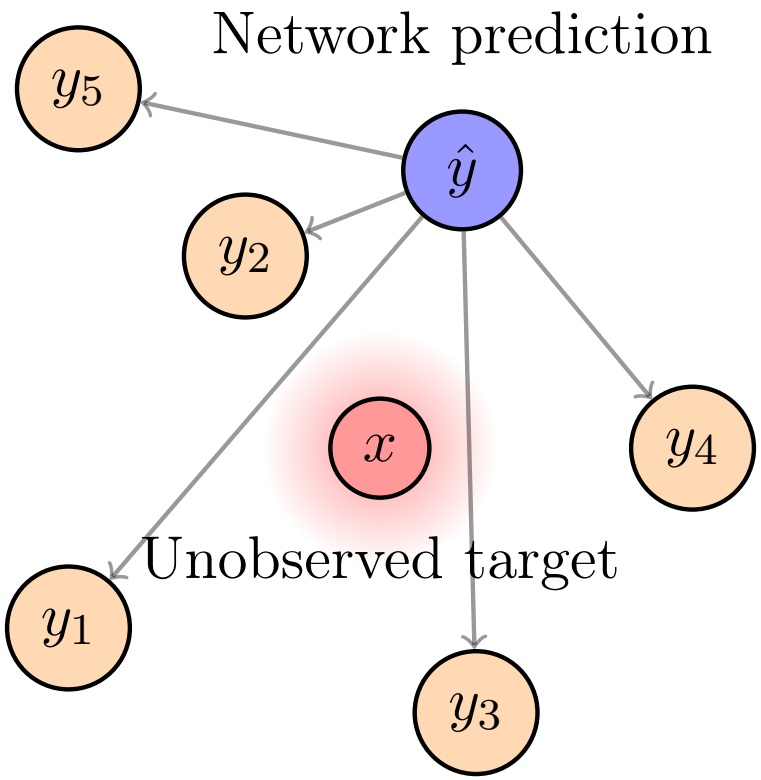
If ρ is the L_1 loss, then proposition 2.1 is satisfied and the optimal solution is the ***median*** *of noise realisations.*

Proposition 2.4

If noise is poissonian ($Y \mid X{=}x \sim \mathcal{P}(x)$) and ρ is the ***Poisson Negative Log-Likelihood*** *($\rho(y^{(1)}, y^{(2)}) = \sum_i y_i^{(1)} \log(y_i^{(2)}) - y_i^{(2)}$), then proposition 2.1 is satisfied.*

Further intuition

Noise2Noise : restore image without clean data



Binomial sampling

Noise2Noise : restore image without clean data

Proposition 2.5

$Y|X$ is 2-divisible.
For instance, $Y_i|X = x \sim Binomial(Ax, \frac{1}{2})$ works and :

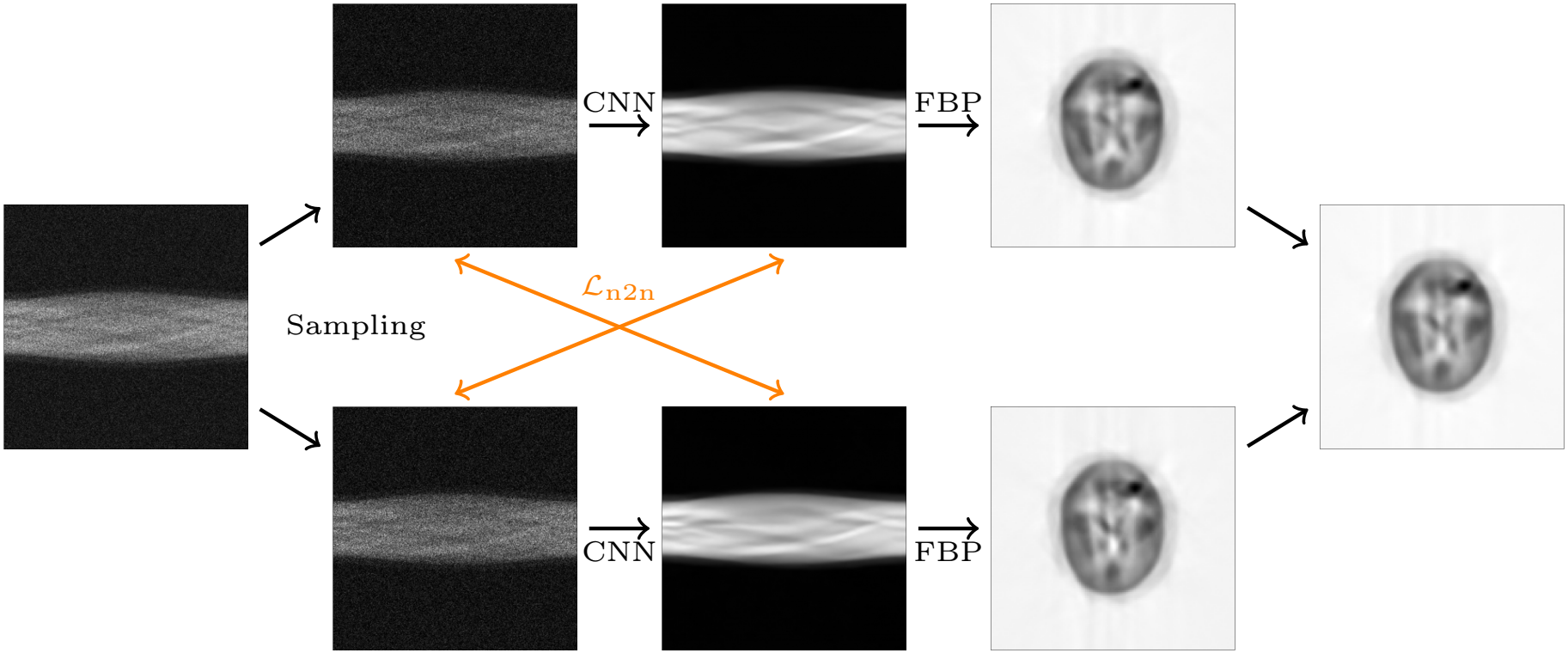
$$\begin{cases} Y_i|X \sim Poisson(\frac{1}{2}Ax) \\ Y_1|X \perp\!\!\!\perp Y_2|X. \end{cases} \tag{5}$$

Hence, statistical conditions 4 are satisfied and Noise2Noise can be applied.

In the Noise2Noise world, this is called "Recorrupted2Recorrupted" [Pang et al. 2021].

Denoising in count domain

Applications



Denoising in count domain

Applications

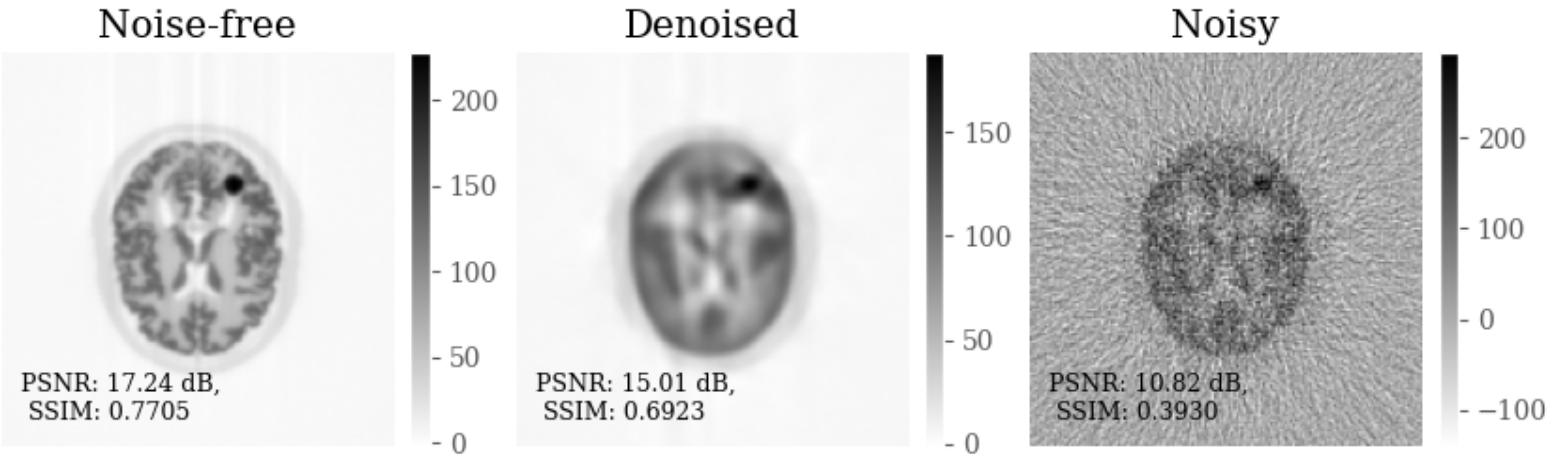
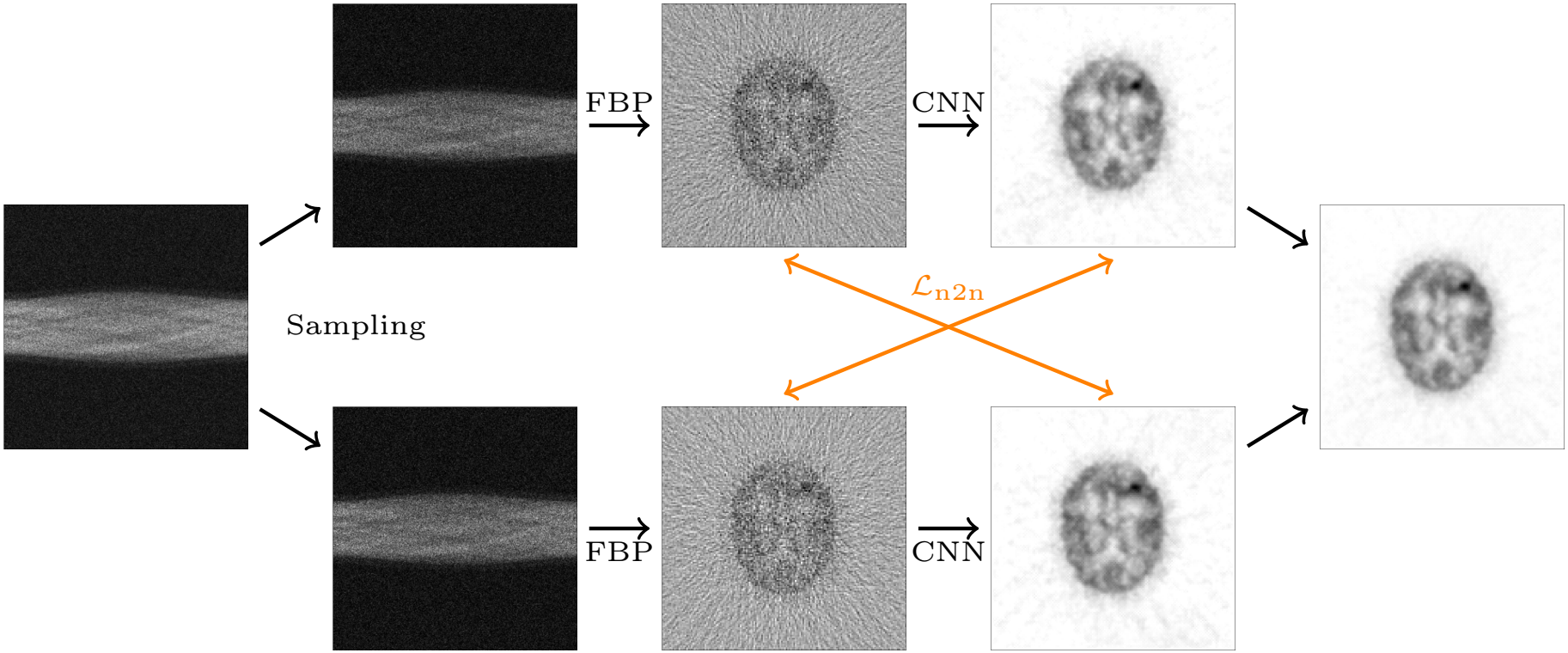


Figure: Photon domain training results

Denoising in image domain

Applications



Denoising in count domain

Applications

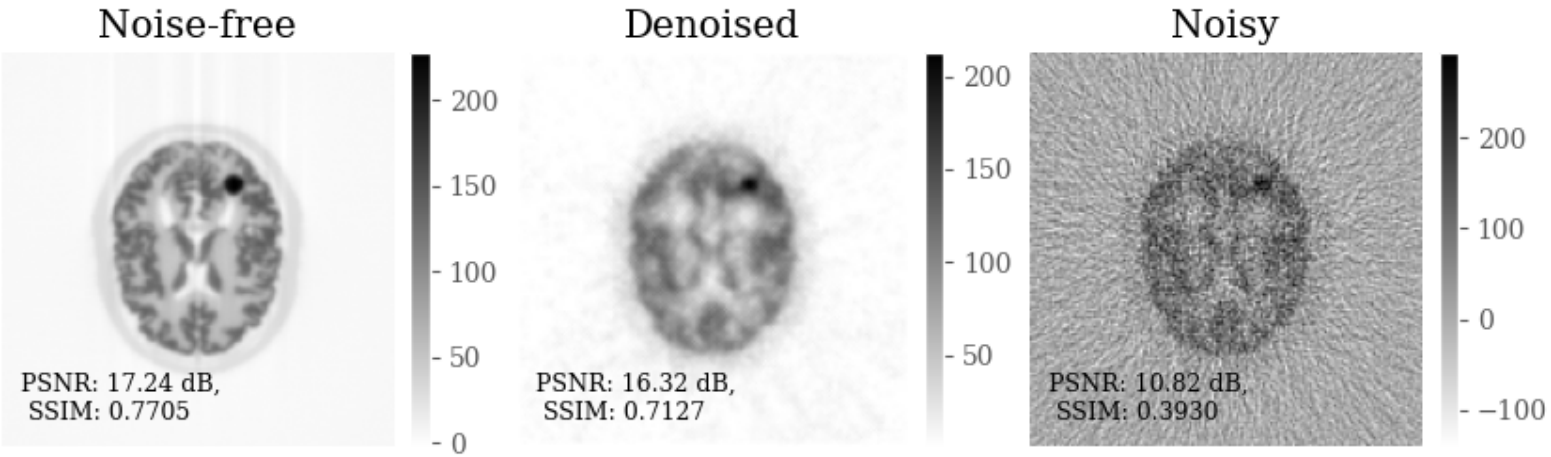
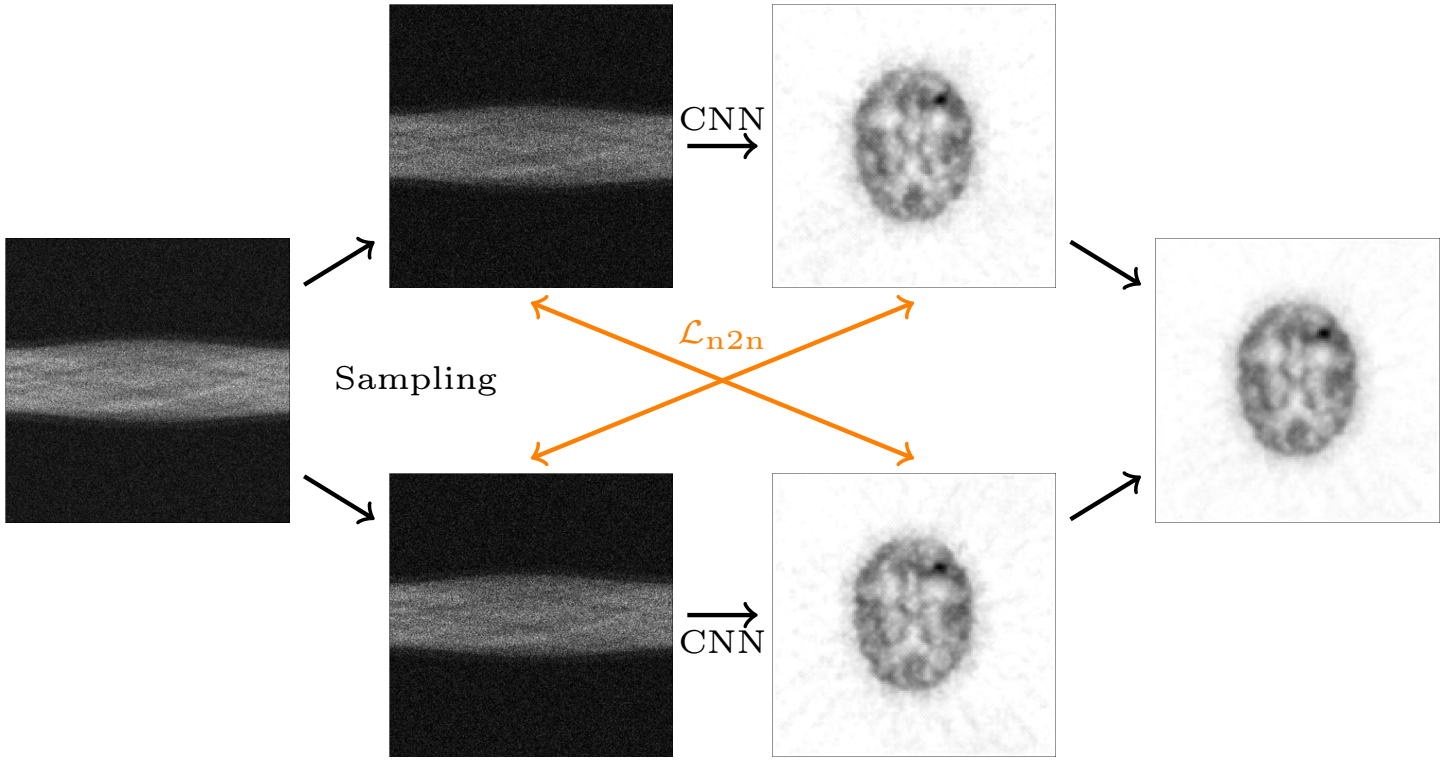


Figure: Photon domain training results

Joint denoising and reconstruction

Application to PET reconstruction



Joint denoising and reconstruction

Work in Progress.

References

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